A STUDY OF THE ECONOMIC IMPACT OF COVID-19 ON THE LIVELIHOOD OF RICKSHAW-PULLERS IN DHAKA



RESEARCH QUESTION: TO WHAT EXTENT HAS COVID-19 IMPACTED THE LIVELIHOOD OF RICKSHAW PULLERS IN DHAKA, BANGLADESH?

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INTRODUCTION

The current global pandemic caused by the Coronavirus Disease 2019 (Covid-19) has led to major economic changes, although mostly adverse, in many countries including Bangladesh and its capital city Dhaka. Similar to many other city structures in developing countries, Dhaka also has an urban informal sector within its very walls. A rickshaw is a man-powered, three-wheeled mode of transport in Bangladesh, that seats one to two passengers for short to mid-distance travel (refer to appendix for a photo). These are generally operated by informal workers of the economy who earn meagre amounts of income.

In 2020 alone during the pandemic, we have seen almost all economic sectors suffer from unemployment, due to extreme risk of contracting the virus, making economic activities less productive and efficient. As rickshaw-pullers are out amidst the public everyday, in contact with other people, they are at high risk from the transmission of, as well as transmitting the disease themselves. This has brought upon less customers every day. With lower odds of earning an income and that of protecting themselves from Covid-19, leading a normal life has become a challenging issue.

There have been all sorts of health, micro and macro-economic impacts of the pandemic in Dhaka. As a Bangladeshi citizen, I have decided to delve deeper into **the economic impact of Covid-19 on the livelihood of rickshaw-pullers in Dhaka**, by giving relative information on the struggles surrounding the livelihood of rickshaw-pullers and how the informal sector workers are generally ignored during economic or health crises**.** Rickshaw-pullers are part of the lower-class and struggled to survive even before the pandemic. Hence, the research question "**To what extent has Covid-19 impacted the livelihood of rickshaw-pullers in Dhaka**?" arose.

HYPOTHESIS

It is likely that the rickshaw-pullers who already suffer from poverty have been adversely affected by Covid-19. The placement of lockdowns, and less provisions of relief from the government may only have worsened the situation. My hypothesis therefore, is that they have been adversely affected significantly.

RESEARCH METHODOLOGY

Primary research was done by interviewing and collecting information from 90 rickshaw-pullers using a questionnaire in certain areas of Dhaka (Dhanmondi, Malibagh and Mirpur) where population and usage of manual transport is high to better understand their financial situation, prospective shock from Covid-19, health concerns and mental well-being. A regression model

will be used to analyse the monthly income and expense changes of rickshaw-pullers due to Covid-19.

Moreover, secondary research will be done from news articles, academic papers with similar topics on the government's methods and policies as well as different society's efforts in tackling Covid-19 and assisting the local rickshaw-pullers. Statistical data from the Bangladesh Institute of Labor Studies, Needs Assessment Working Group Bangladesh, World Bank, BBS Labor Force Survey etc. should help to shed some light on the livelihood of the informal sector amidst this pandemic and the overall existing poverty trap.

DATA AND ANALYSIS

1. Income and expenses of rickshaw-pullers

The best way to understand the impact of Covid-19 on the livelihood of rickshaw-pullers would be to conduct direct interviews with them. The entire interview has been based on 3 specific areas in Dhaka city, that were chosen due to the relatively high volume of rickshaw-pullers and the high probability of usage of rickshaws by the general public. Acquiring the data was not an easy task since Bangladesh is still under restrictions due to the pandemic and a direct interview is the only way to interact with them.

Table 1: Working area of interviewed rickshaw-pullers

Query	Classification	Frequency (N=90)
Working area	Dhanmondi	30 (33.3%)
	Malibagh	30 (33.3%)
	Mirpur	30 (33.3%)

This data is significant given the context of my study because a large volume of rickshaw-pullers reduces the uncertainty and makes the data set more reliable to identify the impact of Covid-19 on the income of a wide range of rickshaw-pullers.

Rickshaw-pullers, like many others in the informal sector, belong to the lower or lower-middle income group. They are usually exempt from public policy making, rights and social protection. Moreover, they face poverty in these modern city areas and are vulnerable to economic shocks and exploitations, such as the one from Covid-19. The idea is that subjectivity makes certain matters, such as the human behaviour towards the informal sector, a matter of opinion which should not be the case with such crucial aspects of human life and occupations. Rickshaw-pullers require a very low skill-set and there are over a million rickshaws in Dhaka alone hence, making up a large part of the urban informal sector (M. Al-Masum. (2019, July 7). Many rickshaw-

pullers do not own rickshaws, and rather hire them on rent, paying an average of 113 BDT per day that do not require any written contract or security deposit (Md. K. Rezaul and S. Khandoker. (2019, March). The minimum fare for a ride is about 25-30 BDT to 100-120 BDT to cover a distance of a 3-kilometer journey in Dhaka, as gathered from the interviews. However, these amounts have fallen due to Covid-19 and have decreased their daily and monthly average income. Income is one of the key indicators to understand the standard of living of an individual and it therefore, can be assumed to have hampered their livelihood for the worse.

Table 2: Income change of rickshaw-pullers in Dhaka (BDT)

Query	Before Covid-19	After Covid-19	Frequency (N=90)
Daily average income	300-450	280-320	21 (23.3%)
	500-750	430-680	47 (52.2%)
	800-1000	700-820	22 (24.4%)
Monthly average income	4000-5000	3500-3800	33 (36.7%)
	5500-6500	4200-5300	45 (50%)
	7000-8000	5500-6800	12 (13.3%)

Table 2 suggests that the daily average income of rickshaw-pullers has fallen as a result of Covid-19 in all frequencies, due to the restrictions in place by the government to increase social distancing and avoid the risk of spread of the disease. When travel is restricted and citizens are advised to reduce interaction with each other, they will generally avoid the usage of public transports, especially rickshaws. However, the poorest members of the society like rickshaw-pullers find themselves unable to adhere to social distancing to survive (I. Sufia, I. Rizwanul, M. Fouzia, R. Sabera and I. Tahiya. (2020, Dec). Using a rickshaw would mean the payment of fare by cash/hand, which increases the risk of transmission of the disease. Therefore, this lowers demand of using the transport. To add to their misery, rickshaw-pullers are also stopped by police officers on roads and restricted to travel to certain areas during certain hours of the day (R. Mizanur. (2020, April 1). Most rickshaw-pullers are married and have at least one to several children who are dependent upon their income. These factors lower their average daily income and thus their monthly average income.

Figure 1: Fall in demand for rickshaw services due to the pandemic

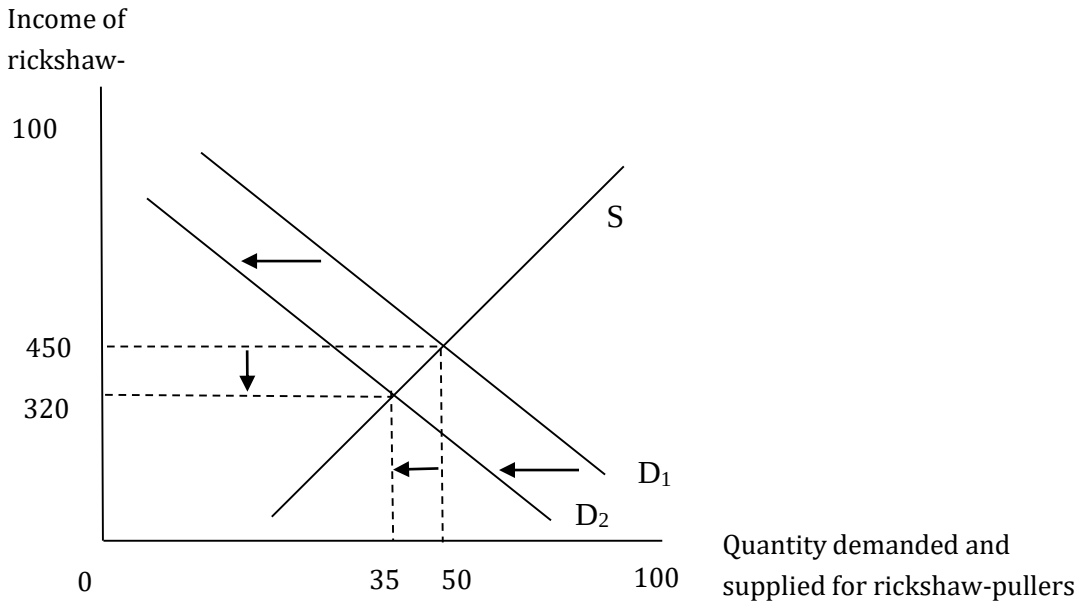


Figure 1 depicts the decrease in demand for rickshaw services due to the pandemic, using the frequency of 21 rickshaw pullers from the total 90 and their maximum daily income from before and after Covid-19 from Table 1. Here, we can assume that the total demand for rickshaw pullers is at a total of 100%. This is because an accurate amount of demand for all rickshaw-pullers cannot be obtained due to lack of time, resources and Covid-19 restrictions. In the figure, the initial demand can be assumed to be at 50% with a maximum daily income of BDT 450 before Covid-19 (D_1). After Covid-19, the demand for rickshaw pullers decreased by about 35%, in return for a maximum daily income of BDT 320 (D_2), as the general population reduced consuming such transport services amidst the lockdown, to top their safety concerns. The shift in the demand curve $D_1 \rightarrow D_2$ also led to a contraction in supply, meaning, how some may have ceased looking for passengers at all, all meaning that their income eventually falls one way or the other. It can be assumed that this is apparent in different areas of the city for most rickshaw-pullers given my broad sampling.

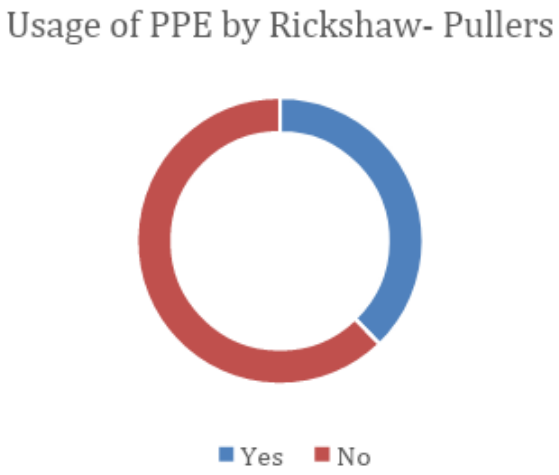
Rickshaw-pullers are becoming more conscious about their daily expenses with an average family size of 4.8 (S. Rubaiyath, H. Masroor, M. Nafees, U. Aftab and M. Parvez. (2020). Their purchasing power has fallen, along with other consumers in the economy due to Covid-19, resulting in many of them temporarily moving back to their homes in rural areas due to the rising expenses in Dhaka and lacking demand.

Table 3: Expense change of rickshaw-pullers in Dhaka (BDT)

Query	Before Covid-19	After Covid-19	Frequency (N=90)
Daily average expenses	50-100	30-70	33 (36.7 %)
	150-200	80-120	35 (38.9%)
	250-300	150-200	22 (24.4%)
Monthly average expenses	1500-2500	1100-1500	27 (30%)
	3000-4000	1800-2400	46 (51.1%)
	4500-5500	2500-3200	17 (18.9%)

The data above has been gathered through the interviews. Table 3 shows that the daily and monthly average expenses of rickshaw-pullers have fallen due to Covid-19. This is because rickshaw-pullers have become more careful of what they spend their income on, as the odds of being able to earn have fallen. Their expenses are usually used to purchase groceries, medicine, pay debts, rickshaw service cost, rickshaw rent, house rent, children's education and other miscellaneous reasons. However, they were reluctant to get into the nit bits of their list of every daily/monthly expense.

Figure 2: Usage of PPE by rickshaw-pullers



Due to Covid-19, rickshaw-pullers have to purchase their own personal protective equipment (PPE) such as hand sanitisers, face masks and gloves as no donations of these items are viable in many areas. Rickshaw-pullers still view these equipment as being not completely necessary, and therefore, do not purchase these items themselves, being reluctant to use them unless they are given and/or forced to wear them in certain areas. During times of crises, such as the pandemic, the government is liable to provide relief to low-income vulnerable groups. However, the poor have to queue up in order to receive the offered aid or be able to purchase groceries at a low cost (I. Sufia, I. Rizwanul, M. Fouzia, R. Sabera and I. Tahiya. (2020, Dec). If their expenses are low, this may mean that they end up eating less and/or acquiring fewer or no medicine which can

negatively affect their health. The impact is vital on their livelihood as the opportunity cost is spending on clothing or other recreational activities.

Table 4: Average changes in the monthly income and expenses of 90 rickshaw-pullers

Month	Income (BDT)	Expenses (BDT)
October (2020)	6000	3500
November (2020)	5400	3100
December (2020)	5000	2800
January (2021)	4500	2500
February (2021)	4700	2600
March (2021)	4600	2400

The information above has been acquired through the interviews. Table 4 portrays the average income and expenditure of the interviewed 90 rickshaw-pullers. It has been accounted for by finding the mean of the range of income and expenditure from the varied frequencies. The data also accounts for 6 months from October 2020, from when the effects of restrictions of Covid-19 was heavy on the rickshaw-pullers, to March, 2021. As Covid-19 was more prevalent during October till December of 2020, most rickshaw-pullers were more adamant on talking about their struggles and changes with income and expenses about that time period. Typically, the average consumer would spend less of their disposable income on expenses as that income declines over a period of time. For example, Table 4 shows their income falling from BDT 6000 in October 2020 to BDT 4000 in March 2021 as well as their expenses declining from BDT 3500 to BDT 2400 over the same time frame.

Figure 3: Regression model and data of monthly average change in income and expenses of rickshaw-pullers during covid-19

SUMMARY OUTPUT	
Regression Statistics	
Multiple R	0.990387832
R Square	0.980868058
Adjusted R Square	0.976085072
Standard Error	64.44552588
Observations	6

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-795.9677419	253.6397698	-3.138181929	0.034908816	-1500.184639	-91.7508447	-1500.184639	-91.7508447
Income (BDT)	0.717741935	0.050120173	14.32042016	0.000138147	0.578586027	0.856897844	0.578586027	0.856897844



The data used for the regression model is a primary source material (interviews of the rickshaw-pullers). The Multiple R. is the correlation coefficient and measures the strength of a linear relationship between two variables: income – independent variable (x-axis), and expenditure – dependent variable (y-axis). The correlation coefficient is closer to a value of 1, meaning the variables have a strong positive relationship. This proves that during the pandemic, as the income of rickshaw-pullers fell, their expenses fell as well. This is a significant aspect of the study as this means that the lower income will ultimately cause a huge impact on their lifestyle and subsistence since they can no longer generate discretionary expenses. R Square. is the coefficient of determination, which is used as an indicator of the goodness-of-fit. Here, the R^2 value is 0.98, which is a fairly good value and means that 98% of the values fit the model. The Standard Error is another goodness-of-fit measure that shows the precision of the regression analysis. The error is 64.4, which is relatively smaller compared to income and expenditure which makes the data set more precise and reliable. A linear regression equation can be created with the income coefficient using the formula:

$y = bx + a$

For this instance,

Y = Income coefficient*x + Intercept

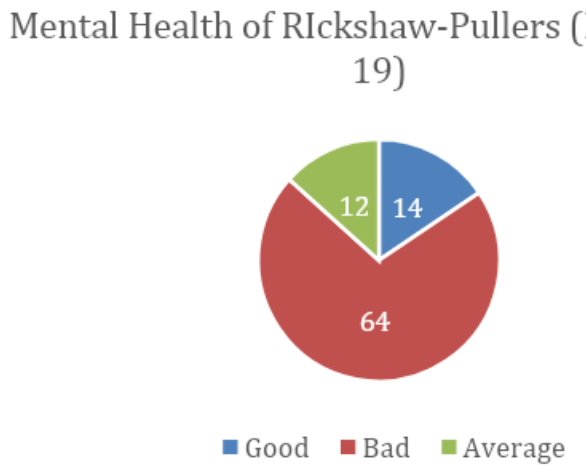
With the values from the regression model,

Y = 0.71*x – 795.867

2. Physical and mental health of rickshaw-pullers

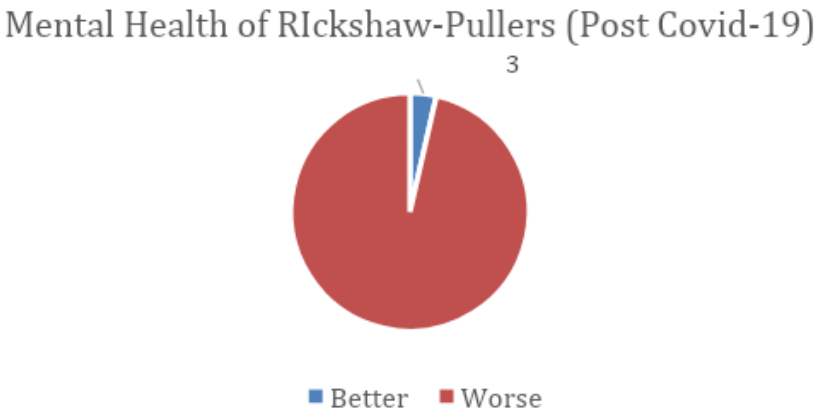
From the information gathered through interviewing the rickshaw-pullers, I have come to understand that the mental and physical health of informal workers was already in a deteriorated state, even before the pandemic. Pulling passengers every day in harsh weather – rickshaw-pulling is a strenuous job. It takes a massive toll on their body. Covid-19 has further degraded their physical and mental health. The ones living in poverty or near the poverty line are more prone to be negatively affected in terms of their mental and physical well-being the earliest (S. Holger. (2018, Jan). This is because they either had the choice of restricting movement and thus suffer from hunger or go to work thereby risk getting infected by the virus. This has increased the level of anxiety, depressive disorders and fear among the rickshaw-pullers (A. Faria & S. Ridwan. (2020, Nov 13).

Figure 4: Mental health of Rickshaw-pullers in Dhaka (Pre-Covid)



*Data obtained from primary data interviews of rickshaw-pullers

Figure 5: Mental health of rickshaw-pullers (post-covid)



*Data obtained from primary data interviews of rickshaw-pullers

The above figures show that the overall mental health of the 90 interviewed rickshaw-pullers was mostly bad to begin with, and that it became even worse for the majority after the pandemic. This is due to vulnerability threats of Covid-19 to their livelihood – the loss of income and the inability to cover expenses. With lower travel demand, rickshaw-pullers have to work overtime to try and earn their daily average whenever and wherever they can, even if that means missing meals, being unable to stay hydrated and/or have ample sleep along the process. Restrictions and lockdowns have also closed down cheap hotels and restaurants where rickshaw-pullers usually dine in, meaning further deterioration of their physical health. Additionally, the fear of economic and food shortage during this pandemic has increased their stress levels. This can be linked to their preferences in economic actions linked to their current emotions (B. Nina. 2009, April 25). Rickshaw-pullers are in a state of panic and excess stress which leads them to consume less and be less responsive and attentive to their overall health.

However, not all the causes of the pressure on their mental and physical health could be accounted for. This is only the information gained which is relevant to Covid-19. The causes could not be determined by one interview, as most rickshaw-pullers were unwilling to share their health issues with a stranger. This is backed up by the fact that the emotional intelligence of rickshaw-pullers is relatively lower than that of an average individual in the economy. They perceive the crisis with the worst possible scenario as they usually do not view such circumstances to have any positive outlook on their lives. This is due to the nature of their livelihood in general and the experiences of being a rickshaw-puller in Dhaka that is embedded into their subconscious, regardless of the pandemic. Human thought is largely made up from images and if all that the rickshaw-pullers see in their surrounding environment are factors that lower their hope, it gravely harms their mental health (B. Nina. 2009, April 25).

Table 5: Family living situation of rickshaw-pullers

Query	Classification	Frequency (N=90)
Living with family or alone	Stays with family	36 (40%)
	Family in rural area	44 (48.9%)

In most households, the rickshaw-puller is usually the only income-earner and has to bear the cost of accommodation. Table 5 shows that 36 of the rickshaw-pullers have their family living with them in Dhaka. Even if their family does not live with them, they still suffer the burden of earning enough to send money back to the village. This puts pressure on them to earn a bare minimum before returning home since if not, they might not be able to afford a home by the end of the month and turn homeless instead.

Table 6: Effects and awareness of Covid-19 of rickshaw-pullers

Query	Classification	Frequency
Aware of Covid-19 and effect	Yes	79 (87.8%)
	No	11 (12.2%)
Aware of ways to prevent	Yes	57 (63.3%)
	No	33 (36.7%)

Likewise, the above data has been gathered from the interviews. Table 6 outlines that even if there are more rickshaw-pullers aware of Covid-19 and its effects, 33 of the 90 interviewed rickshaw-pullers are unaware of the ways of prevention of its transmission. A significant proportion of the people in Bangladesh had poor knowledge about the transmission, signs and symptoms and the incubation period of the Covid-19 virus (K.M. Farhana, K.A. Mannan. 2020). Most of them are unable/unwilling to conduct a Covid-19 test because they are afraid of the expenses of the test as it will ultimately lower their net income. They are unable to focus on any parameter that would allow them to improve their utility in life. Their focus thus, is on objectives that would improve their chances of survivability in the long run, regardless of the opportunity cost of well-being in the short run. This ultimately hampers their livelihood.

3. Poverty trap during Covid-19

Rickshaw pullers are daily earners. When lockdown was imposed, they were not allowed to ride for a long time and acquired less passengers (not to mention being brutally beaten up by the police). All of this enhanced further poverty. Their food supply was tremendously low and instances of corruption arose even in this case. A number of local leaders of the ruling party, Awami League, and local government officials have been arrested for the alleged theft of food items meant for rickshaw-pullers during the Covid-19 shutdown (UCA News reporter, 2020). The sacks of rice were aimed at helping poor and low-income people were hit hard by the loss of work and income during the shutdown, and this situation caused a backlash on social media and the general public.

Poverty can be analysed by the narrow monetary-based measures which are influenced by a lack of social inclusion, discrimination and inequality through political participation (D. Philip & M. Miguel. (2015, June). Covid-19 has only increased the level of poverty in regards to rickshaw-pullers. Neoclassical theories can be applied here where the wide-ranging recognisable reasons for poverty are out of control of rickshaw-pullers. Informal sector workers are usually exempt from social or private assets and are more prone to declining health, lower capability for higher income and barrier to better livelihood. This is more evident during the pandemic.

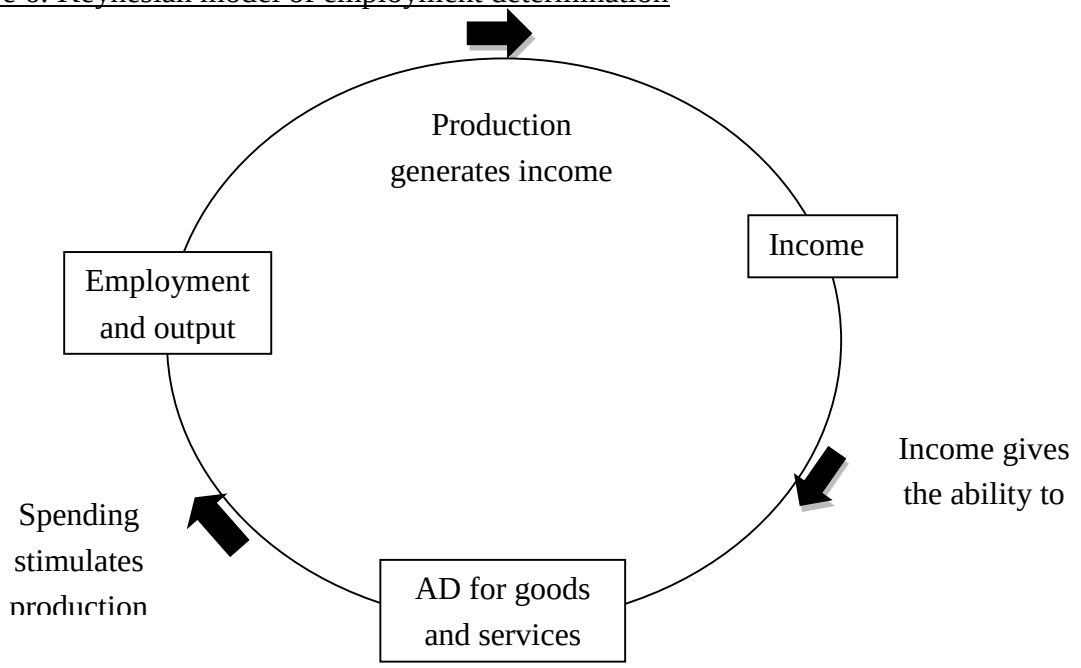
The poverty trap makes people in an economy who are already in poverty being unable to escape from it. It can be created due to a mix of lack of education and healthcare and this phenomenon has further accelerated during Covid-19. The lockdowns imposed have had a crippling effect on workers and businesses across the country (A. Helal. (2020, May 6). The income of informal sector workers is near poverty level due to being unstable. If the poverty line increases, it would create economic powerlessness for millions of workers, leading to a poverty trap. Bangladesh has made progress over the last few years, pulling millions of people up from poverty. Regardless, Covid-19 has been regressing the process.

Table 7: Short-term Poverty Projections under Lockdown.

	Poverty Headcount Rate		
	Baseline (2020Q1)	FY2020Q2	Change
Poverty line = Annual Upper Poverty Line			
Urban	15.8	34.8	19.1
Rural	22.0	27.4	5.4
Total	20.3	29.4	9.1
Poverty line = Annual Lower Poverty Line			
Urban	7.6	33.1	25.5
Rural	14.9	25.9	11.0
Total	10.1	20.5	10.4
Source: Binayak et. al. 2020			

Table 7 depicts that the annual upper and lower poverty lines have both increased for urban and rural areas. About one-fifth of the country's population live below the poverty line and a significant proportion of the workforce is dependent on casual jobs (Sufia, I. Rizwanul, M. Fouzia, R. Sabera and I. Tahiya. (2020, Dec). Among the large proportion of self-employed working people, around 5.19 million self-employed people in the urban informal sector live precariously (R. Islam, 2020). For this instance, let us look at the change in the urban area which has increased considerably during the 2nd quarter of the financial year of 2020. This change can be a cause of the rickshaw-pullers' (of which 1.5 million are part of the urban informal sector) inability to earn the same amount of income that they earned before Covid-19 due to lower passenger demand (Md. K. Rezaul and S. Khandoker, 2019). Increase in stress levels, in turn, has decreased the productivity of rickshaw-pullers. In short, they lack the material means to reduce their own poverty during a pandemic. According to Keynesian theories, rising inequality due to the unequal distribution of income leads to a greater increase in the gap of the poverty line (F. Michael, 2020). Rickshaw-pullers keep on losing opportunities of earning due to Covid-19 restrictions, even after having a relatively lower income compared to the income levels of various other jobs to begin with.

Figure 6: Keynesian model of employment determination

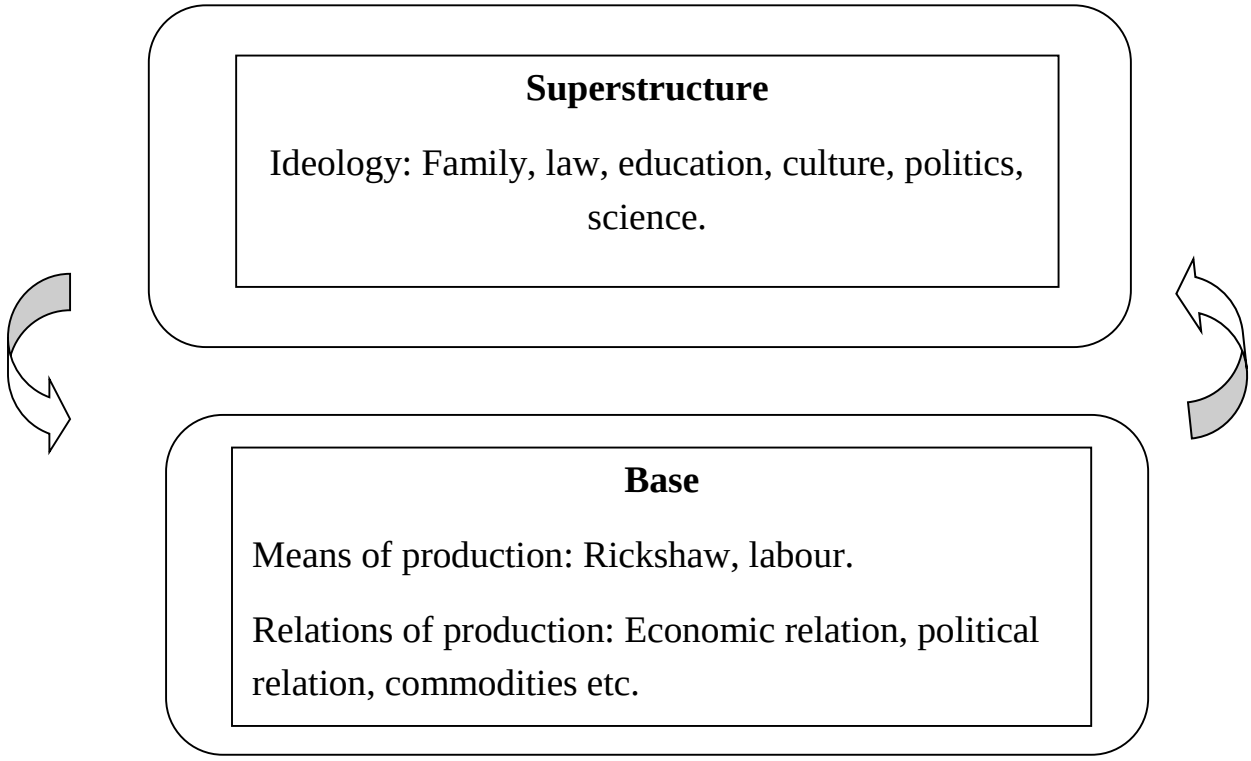


Source: https://www.researchgate.net/figure/The-keynesian-model-of-employment-determination_fig2_325175979

Figure 6 illustrates that income stimulates the ability for consumers to spend on goods and services, which stimulates further employment and output, that can increase production and further improve the ability to increase income in an economy. This is also true in this case with the income earned by rickshaw-pullers, as using that income to consume goods and services aids other businesses and so on – a cycle of growth. The pandemic with all its lockdown measures evidently implies lesser income for rickshaw-pullers to spend on commodities. Moreover, it has also decreased growth and operations of businesses in general, which has further reduced the viability of an average worker to continue to have employment, or work from home. This slowdown of the economy has decreased the circulation of income in the economy and has affected the livelihood of rickshaw-pullers. This has also increased the gap in the poverty line even further for rickshaw-pullers.

Many rickshaw-pullers are also ending up unemployed, or leaving their jobs in the city to find better/alternative opportunities in rural areas. According to Marxist economic theories, the poverty line also focuses on the group characteristics and their status which is dependent on the socio-economic environment which they live in (F. Marriage, 2010).

Figure 7: Marxist base and structure



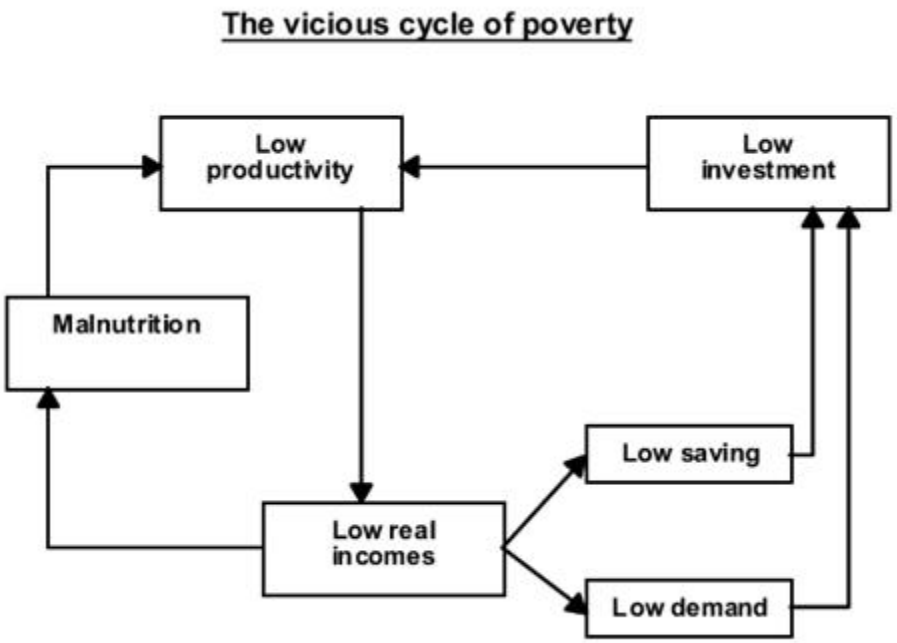
Source: https://literariness.org/2016/04/12/marxism-and-literary-theory/

According to the theory, society can be divided into a base and a superstructure. The base consists of the means and relations of production, while the superstructure consists of the ideologies of society. In some cases, the base is most likely to influence the superstructure while in others, it is the opposite. As rickshaw-pullers are a part of society, their livelihood can be put into the perspective of this diagram and linked to the social consciousness present in society. At present, the circumstances surrounding Covid-19 has been causing changes to their lives in terms of their income and health for the worse. Covid-19 has also changed the way society operates, and has brought up a plethora of barriers for rickshaw-pullers to earn and survive. Thus, we can understand that the superstructure has changed the way people think and has affected their norms and ideologies. The lockdowns, lesser access to passengers and health risks have shifted the base and made their livelihoods worse. Owing to these changes, rickshaw-pullers are more prone to seek help from the government, spend less on commodities and education. This has also led to a greater shift in the poverty line.

The current socio-economic environment in regards to the characteristics of rickshaw-pullers and their livelihood is worse off due to the increase in transmission of Covid-19 (D. Philip & M. Miguel. (2015, June). Rickshaw-pullers are socially excluded from acquiring their basic

necessities when it is required during a pandemic, due to them belonging to the disregarded vulnerable group. However, the projections of the poverty trap can still fluctuate over the coming months as the impact of Covid-19 changes drastically.

Figure 8: Vicious cycle of the Poverty Trap



Source: http://textbook.stpauls.br/Development/page_23.htm

From Figure 8, it can be determined that rickshaw-pullers are part of the vicious cycle of poverty, especially during the pandemic. The demand for rickshaws is lower, which means lower income earning opportunities for them. They are also malnourished due to the physical strain on their body and inability of taking care of themselves. Their productivity is also falling due to mental stress, anxiety and overall depression caused by Covid-19, largely deteriorating their livelihood.

CONCLUSION

"**To what extent has Covid-19 impacted the livelihood of rickshaw-pullers in Dhaka**?" – I have found that, similar to any other economy, as the income of the lower-income class of a country falls, so do their expenses. This is true for the rickshaw-pullers as well, as Covid-19 has decreased passenger demand and hence daily/monthly income. They are also trying to lower their expenses as much as feasible in order to survive. Moreover, I have found that Covid-19 has been putting physical strain on their bodies as they are striving day and night to earn more

income and spend less on food and health products. The mental strain is also evidently present – worries of Covid-19 affecting their family members, ability to pay household expenses and the continuous financial burden of continuing to live in Dhaka city. Even though the government is providing some form of assistance for them, I have found that it has not been enough as they are continuing to struggle as Covid-19 has increased the gap between the rich and poor, decreased opportunities for the lower-income class and further put them deeper into the poverty trap. Therefore, my findings can conclude that Covid-19 has adversely affected the livelihood of rickshaw-pullers in Dhaka significantly.

LIMITATIONS

The limitations of this essay include:

a) **Reliability of certain numeric figures as an exact measure:**
   Some of the information might not have been totally accurate, such as the daily and monthly income and expense, as the rickshaw-pullers were providing it from their memory. The 6-month data for their income and expense figures cannot be trusted to the exact decimal point either since the rickshaw-pullers did not keep strict tabs of them. These can be assumed to be close estimations.
b) **Determining the mental strain on a person:**
   It is not quite possible to fully determine the mental strain of a person without the help from professionals, however, I hope I was able to portray most of it.
c) **I could not find specific primary data from government officials regarding Covid-19 and the impact of it on the livelihood of rickshaw-pullers:**
   In order to compensate for that, I mostly had to use secondary data from news articles instead.

BIBLIOGRAPHY

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APPENDICES

Appendix 1 (Images)

A Bangladeshi Rickshaw



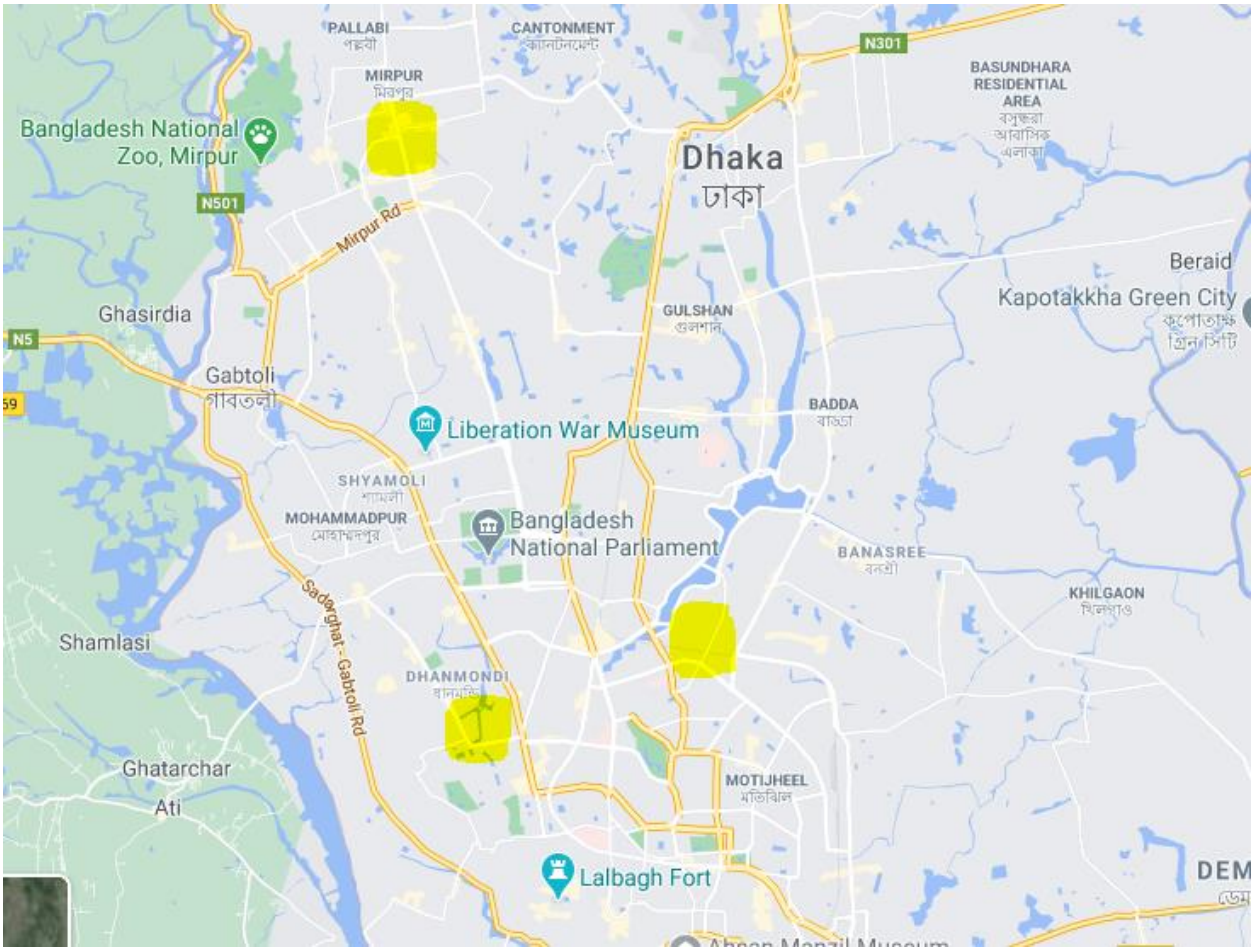
Source: https://sanchitadas22.wordpress.com/tag/bangladeshi-art/

Rickshaw-puller with passenger during Covid-19



Source: https://www.easterneye.biz/bangladesh-readies-150-icu-beds-covid-19-cases-10-now/

Areas of Dhaka city where the interviews have been taken (highlighted in yellow)



Appendix 2 (Interviews)

Interview Questionnaire (Originally taken in Bengali)

1. What is your name?
2. Which area do you usually ride your rickshaw in?
3. How old are you?
4. Are you married?
5. Do you have any children? If so, then how many?
6. Does your family live with you in the city or do they live in the village?
7. Do you own or rent a rickshaw?
8. How much do you have to pay for rickshaw rent?
9. How many days do you work in a week?
10. How long do you work every day?
11. How much do you earn per passenger on average?
12. What was your average daily income before Covid-19?
13. What is your average daily income after Covid-19?
14. What was your average monthly income before Covid-19?
15. What is your average monthly income after Covid-19?
16. How much do you have to pay for the service cost of a rickshaw?
17. How much did you spend on expenses daily before Covid-19?
18. How much do you spend on expenses daily after Covid-19?
19. How much did you spend on expenses monthly before Covid-19?
20. How much do you spend on expenses monthly after Covid-19?
21. What sort of behaviour do you face from the rickshaw owner?
22. What sort of behaviour do you receive from passengers?
23. What sort of behaviour do you receive from the traffic police?
24. How is your mental health?
25. How is your relationship with fellow rickshaw-pullers?
26. Are you aware of Covid-19 and its effects?
27. Do you know about the ways of preventing Covid-19?
28. Do you use proper personal protective equipment?
29. How has Covid-19 affected your mental health?
30. How has Covid-19 affected your income in the past 6 months?
31. How has Covid-19 affected your expenses in the past 6 months?
32. Why do you think your income has changed?
33. Where do you receive your medical services from?
34. Do you take loans from NGOs?
35. Do you take informal loans?